

Isomorphic Open Subsets and Blowing Up

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5 ABSTRACT: This report is a summary on the chapter: “Birational Maps and Blowing Up” from
6 the Algebraic Geometry lecture notes drafted by Andreas Gathman. The main purpose is to recog-
7 nize pairs of varieties that are “almost the same” by developing the necessary language to describe
8 having isomorphic open subsets, in an irreducible setting. We will then show construction (i.e.
9 blowing up) to modify an algebraic variety locally and more importantly, while having open sub-
10 sets unchanged. A couple of concrete examples of blowing up will be presented and the geometric
11 intuition behind the construction will be explained.

1. Introduction

At this point of the Gathmann's lecture notes, we have already seen many examples of varieties: affine varieties, projective varieties, and Grassmanians. Our journey started from the affine space $\mathbb{A}^n$ and common solutions to polynomial equations $V(f_1, \dots, f_r)$ with nullstellensatz that bridged algebra and geometry, and then we moved on to the projective space $\mathbb{P}^n$ and how projective varieties are defined as the common zeros in a homogenous setting. We also familiarized ourselves with a special case of projective varieties that have a very "nice" structure, the Grassmanians $G(k, n)$, which are the sets of all k -dimensional linear subspaces of $\mathbb{P}^n$.

Before moving on to introducing the notion of birational maps and blowing up, we will first look at a couple of examples to motivate the need for these concepts. We will see that there are pairs of varieties that are "almost the same" in the sense that they have isomorphic open subsets, but they are not isomorphic as varieties.

Example 1. Let $X = \mathbb{P}^n \times \mathbb{P}^m$ and $Y = \mathbb{P}^{n+m}$. From a previous exercise, 7.7(c), we know that X and Y are not isomorphic as varieties. However, if we consider the open subsets $\mathbb{A}^n \times \mathbb{A}^m = \mathbb{A}^{n+m} \subseteq X$ and $\mathbb{A}^{n+m} \subseteq Y$, then

$$X \supseteq \mathbb{A}^{n+m} \xrightarrow{\cong} \mathbb{A}^{n+m} \subseteq Y \quad \text{as isomorphic open subsets}$$

Example 2. A more classical example would be let $X = \mathbb{A}^1$ and $Y = V(y^2 - x^3) \subseteq \mathbb{A}^2$ (i.e. a cusp) with the following illustrations:



It is clear that $X \xrightarrow{\neq} Y$: while the map $t \mapsto (t^2, t^3)$ is bijective, its set-theoretic inverse is not a morphism due to the existence of a singularity at the origin so it is impossible to find an inverse

29 morphism. However, if we remove the singularity at the origin, then we have open subsets
30 $\mathbb{A}^1 \setminus \{0\} \subseteq \mathbb{A}^1$ and $V(y^2 - x^3) \setminus \{(0,0)\} \subseteq \mathbb{A}^2$, then the open subsets $X \setminus \{0\} \xrightarrow{\cong} Y \setminus \{(0,0)\}$ are
31 isomorphic under the same map $t \mapsto (t^2, t^3)$ with inverse map $(x, y) \mapsto y/x$.

32

33 Hopefully, these examples have motivated the need for this notion of having isomorphic open sub-
34 sets and the idea of “nearly the same” varieties. The goal of this report is to first develop the neces-
35 sary language to describe having isomorphic open subsets, in an irreducible setting. We will then
36 show construction (i.e. blowing up) to modify an algebraic variety locally and more importantly,
37 while having open subsets unchanged.

38 2. Birational Maps

39 Unless otherwise stated, the varieties of our concern are irreducible. Before introducing the
40 notion of birational maps, we will first define rational maps and rational functions that motivate the
41 former. We will keep referring to the following topological fact over and over both explicitly and
42 implicitly: any nonempty open subset of an irreducible variety is dense in the variety.

43 **Definition 3 (Rational Map)** We denote the rational map from X to Y as $f : X \dashrightarrow Y$ if there exists
44 an open subset $\emptyset \neq U \subseteq X$ such that $f : U \rightarrow Y$ is a morphism.

45 Note that this definition aligns with the idea of “nearly the same”: rational maps do not have to be
46 defined everywhere on X but nonempty open subsets of X , which are dense in X . An equivalence
47 relation naturally arise from this definition. Let $f_1, f_2 : X \dashrightarrow Y$ be two rational maps defined on
48 $U_1, U_2 \subseteq X$ open, respectively, then they are equivalent if they agree on a nonempty open subset of
49 $U_1 \cap U_2$, i.e.

$$f_1 \sim f_2 \iff \exists \emptyset \neq U \subseteq U_1 \cap U_2 \text{ such that } f_1|_U = f_2|_U$$

50 **Example 4.** Consider

$$\begin{aligned} f : U = \mathbb{A}^1 \setminus \{0\} &\longrightarrow \mathbb{A}^1 \\ x &\longmapsto \frac{1}{x} \end{aligned}$$

51 which defines a rational map $f : \mathbb{A}^1 \dashrightarrow \mathbb{A}^1$. Clearly, the following map f' defines the same rational
 52 map:

$$\begin{aligned} f \sim f' : U' = \mathbb{A}^1 \setminus \{0, 1\} &\longrightarrow \mathbb{A}^1 \\ x &\longmapsto \frac{1}{x} \end{aligned}$$

53 since $f|_{U'} = f'|_{U'}$.

54 The following lemma is a direct consequence and stronger statement of the definition of rational
 55 maps and the equivalence relation we just defined.

Lemma 5 *If $f_1 : U_1 \subseteq X \rightarrow Y$ and $f_2 : U_2 \subseteq X \rightarrow Y$ agree on $\emptyset \neq U \subseteq U_1 \cap U_2$, then*

$$f_1|_{U_1 \cap U_2} = f_2|_{U_1 \cap U_2}$$

56 *Proof.* Let $E := \{x \in U_1 \cap U_2 : f_1(x) = f_2(x)\}$, it is clear that E is closed as the preimage of the
 57 diagonal Δ_Y under $(f_1, f_2) : U_1 \cap U_2 \rightarrow Y \times Y$ in $U_1 \cap U_2$. Because $f_1|_U = f_2|_U$, we have $U \subseteq E$.
 58 Since U is nonempty and open, it is dense in $U_1 \cap U_2$, so $\bar{U} = U_1 \cap U_2$. However, $U \subseteq E$ and E is
 59 closed in $U_1 \cap U_2$, so $\bar{U} \subseteq E \subseteq U_1 \cap U_2$, which leads to $E = U_1 \cap U_2$. Therefore, $f_1|_{U_1 \cap U_2} = f_2|_{U_1 \cap U_2}$.
 60 □

61 There exists an issue with the definition of rational maps: it is not clear whether the composition of
 62 two rational maps is well-defined. Consider $g(x) = \frac{1}{x}$, which is geometrically meaningful, but not
 63 everywhere defined on $\mathbb{A}^1$. If $f : \mathbb{A}^1 \rightarrow \mathbb{A}^1$ is a morphism whose image is precisely $\{0\}$, then $g \circ f$
 64 is not well-defined as a rational map because there is no nonempty open subset on which $g \circ f$ is de-
 65 fined. This issue motivates the following definition of dominant rational maps and birational maps.

66
 67 **Definition 6 (Dominant)** *X, Y irreducible varieties. A rational map $f : X \dashrightarrow Y$ is dominant if*
 68 *$\text{Im}(f)$ contains a nonempty open (i.e. dense) subset of Y .*

69 Now let $f : X \dashrightarrow Y$ and $g : Y \dashrightarrow Z$ be rational maps. Suppose that f is dominant, and let $U \subseteq X$ and
 70 $V \subseteq Y$ be nonempty open subsets on which f and g are defined, respectively. Since f is dominant,
 71 the image $f(U)$ contains a nonempty open subset of Y , so in particular $f(U) \cap V \neq \emptyset$. It follows
 72 that

$$f^{-1}(V) \subseteq U$$

73 is a nonempty open subset of U , hence also a nonempty open subset of X . On this subset, the
 74 composition $g \circ f$ is defined. Therefore, $g \circ f : X \dashrightarrow Z$ is again a rational map.

75

76 **Definition 7 (Birational)** $f : X \dashrightarrow Y$ is birational if f is dominant and there exists a rational map
 77 $g : Y \dashrightarrow X$ such that $g \circ f = id_X$ and $f \circ g = id_Y$ as rational maps. In this case, we say that X and
 78 Y are birational.

79 Note that we started with claiming that morphisms are too rigid to capture the idea of “nearly
 80 the same” varieties, and we introduced rational maps to capture this idea by keeping the map but
 81 ignore the small subsets where the map is not defined. We now arrived at the notion of birational
 82 maps, which allows us to compare varieties up to this “same on a nonempty open set” relation.
 83 The following lemma further illustrates the idea of birationality.

84

Lemma 8 X, Y irreducible varieties. X and Y are birational if and only if they contain isomorphic
 nonempty open subsets

$$X \supseteq U \xrightarrow[f]{\sim} V \subseteq Y$$

Proof. The backward direction is trivial. For the forward direction, let $f : X \dashrightarrow Y$ and $g : Y \dashrightarrow X$
 be birational maps such that $g \circ f = id_X$ and $f \circ g = id_Y$. Let $U \subseteq X$ and $V \subseteq Y$ be nonempty open
 subsets on which f and g are defined, respectively. The construction we need here is to further
 restrict the open sets mentioned to a smaller open set where both f and g are defined and they
 are inverses of each other. We consider $U \cap f^{-1}(V)$ and $V \cap g^{-1}(U)$ and clearly both f and g are
 defined on these sets, hence the composition makes sense as morphisms. Since they are also inverse
 of each other, the map

$$f : U \cap f^{-1}(V) \rightarrow V \cap g^{-1}(U)$$

85 is an isomorphism.

86

□

87 Note that our proof essentially demonstrated the idea mentioned above: we kept the map but
 88 ignored a particular subset of the domain where the composition of map is undefined. For example,
 89 we have f and g which are only known on U and V , respectively. To achieve $g \circ f$, we start in U and

after applying f , we end up in V . This is precisely $U \cap f^{-1}(V)$. The following construction is not directly related to defining blow up, but it is essential to explain why blow up is a birational operation.

Definition 9 (Rational Function) *Let X be an irreducible variety. A rational function on X is a rational map $f : X \dashrightarrow \mathbb{A}^1 = K$ where K is the base field. We denote the set of all rational functions on X as $K(X)$, which is called the function field of X .*

To briefly explain the terms: rational function on X is given by a regular function $\varphi \in \mathcal{O}_X(U)$ for some nonempty open subset $U \subseteq X$. Two such functions $\varphi \in \mathcal{O}_X(U)$ and $\psi \in \mathcal{O}_X(V)$ are considered equivalent if they agree on $U \cap V$. The set of equivalence classes is then denoted by $K(X)$. By defining $\varphi_1 + \varphi_2$ and $\varphi_1 \cdot \varphi_2$ on nonempty $U_1 \cap U_2$ for $\varphi_1 \in \mathcal{O}_X(U_1)$ and $\varphi_2 \in \mathcal{O}_X(U_2)$, it can be clearly seen that $K(X)$ is a field. We will state the following important result without proof.

Lemma 10 *If $U \subset X$ is a nonempty open subset of an irreducible variety X , then $K(U) \cong K(X)$ as fields. In particular, if X and Y are birational, then $K(X) \cong K(Y)$ as function fields.*

3. Blow Ups

The goal of the following construction is that given a variety X , we want to construct a new variety $\tilde{X}$ together with a map $\pi : \tilde{X} \rightarrow X$ such that π is a morphism and only birational under the irreducibility assumption. In other words, we want π to be a birational map. As introduced earlier, the philosophy of this construction is to leave dense open subsets unchanged while modifying X “somewhere else”. Hence, it is important to decide where the modification should happen. The plan is to first define blow up in the easiest case, where X affine and Y cut off by finitely many functions, then we later glue these affine constructions together to get the blow up for arbitrary varieties and arbitrary closed subvarieties.

We also have to be cautious in the sense that our construction should not depend on the choice of generators $f_1, \dots, f_r$ but the ideal they generate: if another set of generators $f'_1, \dots, f'_s$ were chosen, then the resulting blow up should be isomorphic to the one obtained by $f_1, \dots, f_r$. This is crucial because the closed subset Y is encoded by the ideal, not the particular choice of generators.

Construction 11 (Blowing Up). Let $X \subseteq \mathbb{A}^n$ be an affine variety and $f_1, \dots, f_r \in A(X)$. Consider their common zero locus $Y = V(f_1, \dots, f_r) \subseteq X$. This is the subset of our choice where all the chosen functions vanish simultaneously, so if the blow up is supposed to modify X locally along Y , then nothing should change away from Y . We will refer to Y as the center of the blow up.

The set that represents “away from Y ” is simply $U = X \setminus Y = X \setminus V(f_1, \dots, f_r)$ such that not all $f_i(x)$ vanish at once on U . Hence the tuple $(f_1(x), \dots, f_r(x))$ is not a zero vector and gives a well-defined point of projective space $\mathbb{P}^{r-1}$. It induces the following map naturally

$$\begin{aligned} f : U = X \setminus V(f_1, \dots, f_r) &\longrightarrow \mathbb{P}^{r-1} \\ x &\longmapsto (f_1(x) : \dots : f_r(x)) \end{aligned}$$

Therefore, we naturally get a well-defined morphism $f : U \rightarrow \mathbb{P}^{r-1}$, because maps to projective space are given locally by regular functions that do not vanish simultaneously, and simply recall that elements of the coordinate ring are regular. This map is crucial as it encodes the projective ratio of the functions $f_1, \dots, f_r$ at every point away from the center. Hence, as we move a point $x \in U$ closer and closer to the center, the individual polynomial values $f_1(x), f_2(x), \dots, f_r(x)$ all go to zero by continuity. However, the ratio $(f_1(x), \dots, f_r(x)) \in \mathbb{P}^{r-1}$ can remain well-defined and inform us the direction from which x is approaching Y . Now, we wish to encode both the points in U and also its projective direction by forming the following graph:

$$\Gamma_f = \{(x, f(x)) : x \in U\} \underset{\text{(closed)}}{\subset} U \times \mathbb{P}^{r-1} \underset{\text{(open)}}{\subset} X \times \mathbb{P}^{r-1}$$

It is clear that the graph sits inside the product space between X and $\mathbb{P}^{r-1}$, where we are allowed to talk about closure in this topological space. Note that we can’t discuss about the closure of f because it would require extra structure to extend the map to points that are not defined. Note that our goal of constructing blow up is that we want a space $\tilde{X}$ that maps back to all of X including Y and replaces Y with something more refined (i.e. with the projective directions). Therefore, we wish to extend the graph by taking its closure to become a subset over X and not just over $U \subset X$:

$$\overline{\Gamma_f} \subset X \times \mathbb{P}^{r-1}$$

125 Over points $x \in U$, we still have $(x, f(x))$ - so we are not modifying this nonempty open subset of
 126 X ; but over points $x \in Y$, the closure fills in the points over Y by remembering all the limit directions
 127 of the paths that can approach x , which is essentially the data from $\mathbb{P}^{r-1}$ that we replace Y with. We
 128 define the blow up of X at $f_1, \dots, f_r$ to simply be this closure:

$$\text{BL}_{f_1, \dots, f_r}(X) := \tilde{X} := \overline{\Gamma_f}_{(\text{closed})} \subset X \times \mathbb{P}^{r-1}$$

129 Once $\tilde{X}$ is constructed inside $X \times \mathbb{P}^{r-1}$, there is an obvious projection to the first coordinate:

$$\pi: \tilde{X} \longrightarrow X$$

Moreover, this projection map π is birational if X is irreducible. Recall that two irreducible varieties are birational if and only if there exists isomorphic open subsets between them. We first note that π is an isomorphism over U . The graph Γ_f is always isomorphic to its domain U , simply because x can be recovered from $(x, f(x))$ by projecting it onto the first coordinate, hence

$$\pi|_{\Gamma_f}: \Gamma_f \xrightarrow{\cong} U$$

130 Also, U and Γ_f are both dense: U is dense in X due to irreducibility and $U = X \setminus Y$ is a nonempty
 131 open subset under the assumption that $Y \subsetneq X$; Γ_f is dense in $\tilde{X}$ simply by its definition. Now, we
 132 have two dense open subsets $U \subset X$ and $\Gamma_f \subset \tilde{X}$ which are isomorphic to each other. Therefore X
 133 and $\tilde{X}$ are birational.

134

135 To conclude, $\pi: \tilde{X} \longrightarrow X$ is always a morphism and in fact always an isomorphism over U ; when
 136 X is irreducible and $Y \subsetneq X$, π is additionally birational since $U \underset{(\text{dense})}{\subseteq} X$ and $\Gamma_f \underset{(\text{dense})}{\subseteq} \tilde{X}$. Our
 137 construction leads to the following definition very naturally

Definition 12 (Exceptional Set) *According to Construction 11, Γ_f is the part of $\tilde{X}$ that corresponds to the unchanged open set U , hence we denote the modified part as the complement:*

$$E = \tilde{X} \setminus \Gamma_f = \pi^{-1}(V(f_1, \dots, f_r)) \quad (\text{i.e. the exceptional set lies precisely over the center})$$

138 which corresponds to the part newly created by the blow-up.

139 **Construction 13 (Subvarieties and Blowing Up).** Suppose we have a closed subvariety $Z \subset X$,
 140 then we can blow up Z at $f_1, \dots, f_r$ as well (since our construction above started off locally, it
 141 should be clear that blowing up a closed subvariety is intuitive). By property of regular functions,
 142 $f_i \in A(X)$ naturally restrict to $f_i|_Z \in A(Z)$, so the same candidates $f_1, \dots, f_r$ also defines a blow-up
 143 of Z .

144

The first step is to identify the open part that remains the same. Recall that $U = X \setminus Y$ is the part
 of X where the blow-up doesn't modify anything, and $\tilde{X} \supseteq \Gamma_f \cong U$ under π . Now restrict Z inside
 U with $Z \cap U$, such a restriction can be identified as a subset of $\tilde{X}$ under the isomorphism $\Gamma_f \cong U$.
 We then define $\tilde{Z} := \overline{Z \cap U} \subset U \subset \tilde{X}$ in $\tilde{X}$, and by construction:

$$\tilde{Z} =: \text{BL}_{f_1, \dots, f_r}(Z) \subset Z \times \mathbb{P}^{r-1} \subset X \times \mathbb{P}^{r-1}$$

145 which makes $\tilde{Z}$ a closed subvariety of $\tilde{X}$. This closure is referred to as the “strict transform” of Z .

We will briefly discuss the case when X is reducible. Suppose it has the irreducible decomposition
 $X = X_1 \cup \dots \cup X_m$, then intuitively each component carries the same construction, and the blow up
 of X is obtained by blowing up the components individually then take the union:

$$\text{BL}_{f_1, \dots, f_r}(X) = \text{BL}_{f_1, \dots, f_r}(X_1) \cup \dots \cup \text{BL}_{f_1, \dots, f_r}(X_m) = \tilde{X}_1 \cup \dots \cup \tilde{X}_m$$

146 This idea is natural because the whole construction is local, so blowing up the decomposition
 147 consisting of closed pieces are simply repeating Construction 13 for m -times.

148 4. Concrete Examples of Blow Up

149 *a. Trivial Example: $r = 1$*

150 The motivation of this trivial example is to investigate whether the construction of blow up is
 151 meaningful: if it is, then we expect to see only triviality happens in this trivial case. When $r = 1$,
 152 the projective space is $\mathbb{P}^{1-1} = \mathbb{P}^0$, a single point $\{*\}$. Hence: $X \times \mathbb{P}^0 \cong X$, which is already hinting

153 that the construction is very simple in this case.

154

155 Again, let $Y = V(f_1)$ be the locus of zeroes, which is our center. The open subset that remains
 156 unchanged under blow up is $U = X \setminus Y$, the complement of the zero locus. Consider the same map
 157 that maps to the set of lines the origin of a space V where $\dim(V) = 1$:

$$\begin{aligned} f: U = X \setminus V(f_1) &\longrightarrow \mathbb{P}^0 \\ x &\longmapsto * \end{aligned}$$

Note that this map on U to $\mathbb{P}^0$ is fixed: there is only one possible value. Then the graph

$$\Gamma_f = \{(x, f(x)) : x \in U\} = \{(x, *) : x \in U\} \underset{(\text{closed})}{\subset} U \times \mathbb{P}^0 \cong U \underset{(\text{open})}{\subset} X \times \mathbb{P}^0 \cong X$$

is just $U \cong \Gamma_f$ itself inside X . The blow up is then just the closure of the graph:

$$\tilde{X} = \overline{U} \subseteq X$$

158 Therefore, the blow up in the case $r = 1$ is simply the closure of the complement of $V(f_1)$. If X is
 159 irreducible and $f_1 \neq 0$ then $V(f_1)$ is a proper closed subset and U is a nonempty open subset. By
 160 the fact mentioned at the beginning, we can conclude that U is dense and $\tilde{X} = \overline{U} = X$. Then the
 161 blow up is just X itself.

162

163 We might be interested in the case of $f_1 = 0$, then $V(f_1) = X$ and $U = X \setminus V(f_1) = X \setminus X = \emptyset$.
 164 Clearly, the graph is empty and its closure is empty, therefore $\tilde{X} = \emptyset$. This shows that with only one
 165 function, this construction carries nothing geometrically meaningful. It is obvious that meaningful
 166 discussion comes when $r \geq 2$ where projective ratio matters.

167 Note that closure argument in our blow up construction is conceptually clean in the theoretical
 168 sense – but blow up is intrinsically a computation, and it would be difficult to compute the actual
 169 closure of a graph. Hence, we would like to have a more concrete description based on equations.
 170 This next lemma provides us with a tool to replace the abstract closure definition with a closed
 171 subvariety of $X \times \mathbb{P}^{r-1}$ cutoff by equations.

Lemma 14 *The blow-up $\tilde{X}$ of an affine variety X at $f_1, \dots, f_r \in A(X)$ satisfies*

$$\tilde{X} \subset \{(x, y) \in X \times \mathbb{P}^{r-1} : y_i f_j(x) = y_j f_i(x) \quad \forall i, j = 1, \dots, r\}$$

Proof. We provide the intuition and proof skeleton here. The lemma itself is literally stating that the blow up $\tilde{X}$ is contained in the set defined by the equations $y_i f_j(x) = y_j f_i(x)$ for all i, j . First suppose we have two equivalent projective points $(y_1 : \dots : y_r) = (f_1 : \dots : f_r)$, then

$$(y_1 : \dots : y_r) = (f_1 : \dots : f_r) \iff y_i f_j(x) = y_j f_i(x) \quad \forall i, j$$

172 Generally, a point $(x, y) \in X \times \mathbb{P}^{r-1}$ has $y = (y_1 : \dots : y_r)$ as projective coordinates. If we focus on
 173 the graph $\Gamma_f \subset X \times \mathbb{P}^{r-1}$, then any point in Γ_f satisfies $y = (f_1 : \dots : f_r)$ by definition, so the equation
 174 above holds on Γ_f . Besides, the set $\{y_i f_j(x) = y_j f_i(x)\}$ is defined by polynomial equations and
 175 hence is a closed subset of $X \times \mathbb{P}^{r-1}$ by Zariski topology. Since Γ_f is contained in this closed set,
 176 its closure $\tilde{X} = \overline{\Gamma_f}$ must also be contained in it.

177 □

178 *b. Blow up of $\mathbb{A}^n$ at The Coordinate Functions*

179 We first consider the setup: take $X = \mathbb{A}^n$ and $f_i = x_i$ for $i = 1, \dots, n$ (i.e. the linear equations on
 180 each coordinate), then the common zero locus is simply $Y = V(x_1, \dots, x_n) = \{0\}$ and $U = \mathbb{A}^n \setminus \{0\}$,
 181 which is the punctured affine space.

182 By applying the previous lemma, we see that since $f_i = x_i$, the equations become simply

$$y_i x_j = y_j x_i \quad \text{for all } i, j$$

Therefore, the lemma asserts that

$$\tilde{\mathbb{A}}^n \subset Z := \{(x, y) \in \mathbb{A}^n \times \mathbb{P}^{n-1} : y_i x_j = y_j x_i \quad \forall i, j = 1, \dots, n\}$$

Geometrically, the defining equation is saying that $y = (y_1 : \dots : y_n)$ records the direction of $x = (x_1, \dots, x_n)$. When away from the origin, y is just the direction of x ; at the origin, $x = 0$ makes the equation trivial, so y can be anything in $\mathbb{P}^{n-1}$, and this is where the projective directions gets added. Recall that for any blow up, the projection map π is always an isomorphism over $U = X \setminus Y$, so this blow up $\tilde{\mathbb{A}}^n$ is isomorphic to $\mathbb{A}^n$ on the open set U .

In fact, this subset relation is actually an equality. This is worth investigating because it gives us a complete and explicit model of the blow up free from having closure. To prove the following claim, we need to analyse the affine charts of Z : $Z \subset \mathbb{A}^n \times \mathbb{P}^{n-1}$ involves projective space, which isn't affine. However, we can cover $\mathbb{P}^{n-1}$ by affine charts $U_i = \{y_i \neq 0\}$.

Claim. $\tilde{\mathbb{A}}^n = Z$

WLOG, consider $U_1 = \{y_1 \neq 0\}$. In projective space, homogeneity states that

$$(y_1 : \dots : y_n) = \frac{1}{\lambda} (y_1 : \dots : y_n)$$

for any $\lambda \neq 0$. Now because $y_1 \neq 0$, we can choose $\lambda = y_1$ and hence

$$(y_1 : \dots : y_n) = (1 : \frac{y_2}{y_1} : \dots : \frac{y_n}{y_1})$$

We now normalized the projective point by setting the first coordinate to be 1, which turned the projective coordinates into affine coordinates $(y_2, \dots, y_n)$. The equations $y_j x_i = y_i x_j$ for all i, j become

$$y_j x_1 = y_1 x_j \quad \Rightarrow \quad x_j = y_j x_1 \quad \text{for } j = 2, \dots, n$$

194 We observe that all the x_j for $j \geq 2$ are completely determined by x_1 and y_j . The free coordinates
 195 are just $(x_1, y_2, \dots, y_n)$, which gives us an isomorphism $U_1 \cong \mathbb{A}^n$:

$$\begin{aligned} f: \mathbb{A}^n &\xrightarrow{\cong} U_1 \subset \mathbb{A}^n \times \mathbb{P}^{n-1} \\ (x_1, y_2, \dots, y_n) &\longmapsto ((x_1, x_1 y_2, \dots, x_1 y_n), (1 : y_2 : \dots : y_n)) \end{aligned}$$

For the x -coordinate, x_1 is known from the input hence from equations $x_j = y_j x_1$ for $j \geq 2$; for the
 y -coordinate, $y_2, \dots, y_n$ are free since we are on the patch $y_1 = 1$. Note that the same argument
 applies to all other charts U_i , because for $i, j \geq 2$, substituting $x_j = y_j x_i$ and $x_i = y_i x_1$ gives us:

$$y_i x_j = y_i (y_j x_1) = y_j (y_i x_1) = y_j x_i$$

196 Hence the results are consistent with each U_i , and we can conclude that Z is covered by affine
 197 charts U_i which are all isomorphic to $\mathbb{A}^n$, irreducible and of dimension n .

198

199 Now, our goal is to argue that Z is irreducible. Recall that a variety is irreducible if it cannot be
 200 written as a union of two proper closed subsets. BWOC: $Z = A \cup B$ where A, B are proper closed
 201 subsets. Since each U_i is irreducible and isomorphic to $\mathbb{A}^n$, we have either $U_i \subseteq A$ or $U_i \subseteq B$ for
 202 each i because an irreducible space cannot be split between two closed sets. However, the overlaps
 203 force everything to be in the same piece. Suppose $U_1 \subseteq A$, then U_2 cannot be in B because
 204 $U_1 \cap U_2 \neq \emptyset$, so $U_2 \subseteq A$. By the same argument, U_3 would intersect with some U_i already in A ,
 205 so U_3 must also be in A . Continuing, we conclude that all U_i are in A , which means B is empty,
 206 contradiction.

207

208 Therefore, Z can't be split into separate closed pieces, which forces Z to be irreducible. Since
 209 each chart is n -dimensional, and they glue together into one irreducible variety, we conclude that
 210 Z is an irreducible variety of dimension n .

211

212 We now argue that both the blow up and Z agree on the dense open set U . By construction, $\widetilde{\mathbb{A}^n}$
 213 is the closure of Γ_f , and Γ_f is precisely the graph of f on U . On the other hand, Z is defined
 214 by the equations $y_i x_j = y_j x_i$ for all i, j , which are satisfied by points in Γ_f because on Γ_f , we

215 have $y = (f_1 : \dots : f_r) = (x_1 : \dots : x_n)$, so the equations hold. Conversely, for $x \neq 0$, at least one
 216 $x_k \neq 0$, and the equations $y_i x_k = y_k x_i$ then force $y_i = y_k \cdot \frac{x_i}{x_k}$ for all i , meaning $y = (x_1 : \dots : x_n)$ is
 217 uniquely determined. Hence Z over U contains nothing more than Γ_f , and so $\widetilde{\mathbb{A}}^n$ and Z agree on U .

218

Since $\widetilde{\mathbb{A}}^n$ and Z are both irreducible of dimension n and they agree on the dense open set $U = \mathbb{A}^n \setminus \{0\}$, we conclude that

$$\widetilde{\mathbb{A}}^n = \text{BL}_{x_1, \dots, x_n}(\mathbb{A}^n) = \{(x, y) \in \mathbb{A}^n \times \mathbb{P}^{n-1} : y_i x_j = y_j x_i \quad \forall i, j = 1, \dots, n\} = Z$$

Note (Blow Up Morphism). Let us further analyze the blow up morphism $\pi : \widetilde{\mathbb{A}}^n \rightarrow \mathbb{A}^n$ in this case. We first observe that nothing changes when away from the origin: if $x \neq 0$, then the projective coordinates y are determined by the direction of x uniquely. Therefore:

$$\pi^{-1}(\mathbb{A}^n \setminus \{0\}) \cong \mathbb{A}^n \setminus \{0\}$$

The blow up left U unchanged. However, the fiber over the origin is more interesting. At the origin, the equations $y_i x_j = y_j x_i \iff y_i \cdot 0 = y_j \cdot 0$ are trivial, so y can be any point in $\mathbb{P}^{n-1}$. Therefore:

$$\pi^{-1}(\{0\}) \cong \mathbb{P}^{n-1}$$

The blow up replaces the origin with a projective space of dimension $n - 1$, which is the space of directions from which we can approach the origin. By definition, the exceptional set E is just this fiber over the origin:

$$E = \pi^{-1}(\{0\}) \cong \mathbb{P}^{n-1}$$

To conclude, the blow up leaves $\mathbb{A}^n \setminus \{0\}$ unchanged and replaces the origin $\{0\}$ with $\mathbb{P}^{n-1}$. Because the blow up only happened at the origin (i.e. a single point), we can denote the blow up as

$$\text{BL}_0(\mathbb{A}^n)$$

219 This is a new variety of the same dimension as $\mathbb{A}^n$.

220

5. Geometric Intuition

We just showed algebraically that the fibre of $\{0\}$ is $\mathbb{P}^{n-1}$, which is the space of directions from which we can approach the origin. However, we have to check why this makes sense geometrically. Recall that a line through the origin with direction vector $v = (v_1, \dots, v_n) \neq 0$ is

$$L = \{(\lambda v_1, \dots, \lambda v_n) : \lambda \in K\}$$

Two vectors define the same line exactly when they differ by a nonzero scalar multiple, which is precisely the equivalence relation in projective space. Hence, the set of all lines through the origin is naturally $\mathbb{P}^{n-1}$.

Let us consider the strict transform $\tilde{L}$ of L , which is the closure of $L \cap U$ in $\tilde{\mathbb{A}}^n$. Clearly:

$$L \cap U = \{(\lambda v_1, \dots, \lambda v_n) : \lambda \neq 0\}$$

which is the line minus the origin, but each point on this line has the projective direction

$$(v_1 : \dots : v_n) \in \mathbb{P}^{n-1}$$

Hence, the points on $L \cap U$ on $\tilde{\mathbb{A}}^n$ are of the form

$$((\lambda v_1, \dots, \lambda v_n), (v_1 : \dots : v_n))$$

Notice that the projective coordinate is constant $\forall \lambda \neq 0$, so every point on the line has the same direction. When we take the closure, the limit point $((0, \dots, 0), (v_1 : \dots : v_n))$ as $\lambda \rightarrow 0$ is added. Therefore, the strict transform of the line meets the exceptional set $\pi^{-1}(0) \cong \mathbb{P}^{n-1}$ at exactly the single point $(v_1 : \dots : v_n)$. This is why the fibre is essentially the space of directions geometrically.

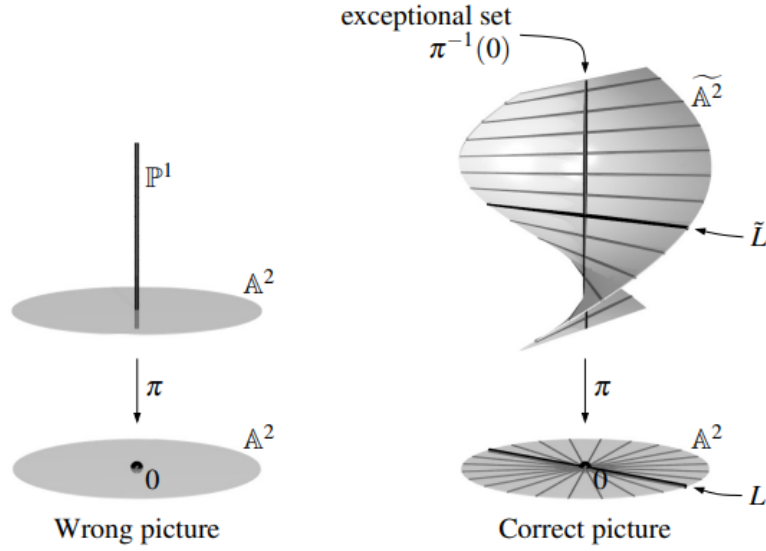


FIG. 1. Geometric Intuition of Blow Up

This graph here essentially captures the geometric intuition of blow up with $\mathbb{A}^2$ as the ambient space. The graph on the left is the wrong picture: $\mathbb{P}^1$ is perceived as a straight line but mistakenly inserted right on top of the origin of $\mathbb{A}^2$. It conveys the idea that exceptional set is just sticking above the origin and signals no relationship with the geometry of $\mathbb{A}^2$. This is wrong in the sense that the exceptional set is being “manually attached” to the origin.

The correct perception is the one on the right. The $\mathbb{A}^2$ at the bottom is the original ambient space, and there are plenty of lines L passing through the origin, each with its own direction. The new space constructed by the blow up is the space above, which is $\tilde{\mathbb{A}}^2$. Every line L at the bottom has its corresponding strict transform $\tilde{L}$ in the blow up. The key is when $\tilde{L}$ approaches the exceptional set, it will land on a unique point in $\pi^{-1}(\{0\}) \cong \mathbb{P}^1$ that corresponds to the direction of L . Hence, it leads to the fan shape of the blow up, which is the union of all the strict transforms $\tilde{L}$ of lines L through the origin. Hence, the exceptional set is never an artificial attachment but rather it “grows” naturally out of the geometry of $\mathbb{A}^2$ by keeping track of the directions of lines through the origin.

6. Conclusion

This example is classic and important, because it demonstrated the useful application of blow up in resolving singularities. Having singularities is a common circumstance in algebraic geometry and is intrinsically local, we usually perceive singularities as a place where “bad” geometry happens. To solve this problematic issue, we replaced this singularity with another geometric object that is more refined by blow up: the projective space of directions. Instead of having all directions collapsing into one single point, we now have a whole space of directions keeping track of each path that our point can approach this singularity separately. Hence, the blow up is a powerful and effective tool in the study of algebraic geometry by smoothing out singularities with these projective directions.

253 *Acknowledgments.*

254 *Data availability statement.*